

4. **Its polyglot nature:** Heroku supports a wide number of programming languages—the popular ones are Ruby, Python, Java, PHP and JavaScript. The comfort of sharing a single deployment platform for a variety of languages is much appreciated by many developers.
5. **Free basic usage:** Every user-deployed application in the Heroku ecosystem has access to a free plan. We will discuss the pricing model soon.

A brief history lesson

Heroku had a humble beginning, graduating from the famous American start-up accelerator, Y Combinator's batch of 2008. Back then, the Ruby on Rails Web framework had started to gain traction, but lacked an easy-to-deploy platform. Heroku established itself as a popular choice among the Rails developer niche. *Salesforce.com* later acquired Heroku in December 2010. Slowly, support for a large number of languages and databases was added. Major tie-ups with Facebook to promote using Heroku for deploying applications and games further strengthened the platform. Heroku also enjoyed the limelight for employing respected developers such as Yukihiro Matsumoto, the creator of the Ruby programming language.

Dynos, stacks and add-ons

A *dyno* is a single instance of an application running on Heroku. Under the hood, a dyno is an Ubuntu server with a Quad Core Processor and 512 MB of RAM. Each dyno has an isolated *chroot* jail environment (implying restriction to a single directory, with no system files access). A dyno can run any type of service such as *Web* (serving HTTP requests), *worker* (asynchronous jobs) and *scheduler* (for scheduling capabilities like *cron*).

The flexibility of Heroku lies in scaling the number of dynos active at a certain point in time. All incoming HTTP requests are passed through a load balancer that routes them to multiple dynos, allowing an application to serve more users concurrently. Heroku's architecture is sure to fascinate any scalability geek. Visit <http://www.heroku.com/how> to delve deeper into the architecture.

The OS, along with language runtime and libraries, makes up a 'Heroku stack' that provides a software environment for the applications. Heroku currently uses an Ubuntu 10.04-based 'Cedar' stack, and an older Debian Lenny 5-based 'Badious Bamboo' stack. However, you cannot use the 'Badious Bamboo' for a new application. You can read more about the Heroku stacks at <https://devcenter.heroku.com/articles/stack>.

Heroku add-ons are great infrastructure services that simplify deployment. To name a few benefits, add-ons provide data stores, caching servers, searching infrastructure, media management and payment gateways. Like dynos, add-ons can be dynamically attached or removed. For the complete add-on list, visit: <https://addons.heroku.com/>.

Pricing

The Heroku pricing model takes a while to get used to. The cost is calculated on 'wall-clock' usage rather than

CPU time, implying usage regardless of the traffic an application receives. Each application receives 750 free dyno-hours per month. Beyond that, a charge of \$ 0.05 per hour is incurred per dyno.

To use an add-on, you need to verify your account first with credit card information. Every add-on has a different pricing model, and many come bundled with a limited free plan as well. To get a more concrete understanding of pricing, visit: <https://devcenter.heroku.com/articles/usage-and-billing>.

The Heroku workflow

Let us walk through the process of creating a Heroku account and deploying a project. Before you can proceed any further, you need to create an account at: <https://api.heroku.com/signup>. After signing up, install the Heroku *toolbelt* command-line utility available for all major platforms—visit <https://toolbelt.heroku.com/> and follow those instructions. Next, log in with the *heroku* command-line utility. You will be asked to enter the email address and password for the account:

```
mythlabs:testapp myth$ heroku login
Enter your Heroku credentials.
Email: nitishupreti@gmail.com
Password (typing will be hidden):
Authentication successful.
```

The next step is to create an application on Heroku, as follows:

```
mythlabs:testapp myth$ heroku create
Creating infinite-reef-6834... done, stack is cedar
http://infinite-reef-6834.herokuapp.com/ | git@heroku.com:infinite-reef-6834.git
```

If you do not specify a name for the application (as in the example above), Heroku comes up with a name for you. You may, however, want to specify a name when executing the create command. Finally, in the directory where the project's Git repository resides, issue *git push heroku master* to deploy your application. This is all you need to deploy your application on Heroku. As advertised earlier, Heroku does truly simplify deployment.

Final thoughts

Heroku is an amazing platform for hackers to quickly deploy scalable Web applications. I hope you have been intrigued by it. It is now time to build something cool and have some fun! Happy hacking! 

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